

Roognis AI — Verified Final HLD

All 8 review fixes applied · Demo-safe · Cloud-portable · Ready for LLD

✔ 8/8 FIXES APPLIED

✔ COMFYUI ADDED

✔ CHROMADB PERSISTENT

✔ RBAC + HTTPONLY JWT

✔ PARENT↔STUDENT RELATION

✔ ANALYTICS EVENT FLOW

✔ RAG CHUNKING DEFINED

🐳 DOCKER-COMPOSE UP → FULL SYSTEM

CLIENTS · BROWSER

NEXT.JS · :3000

Web App — Role-Based Portals

Student: AI chat (SSE stream) · image gen · video playback · feedback

Teacher: batch analytics · query trends · intervention alerts

Parent: child progress · subject scores · attendance

JWT in HttpOnly cookie · 24h expiry

EventSource API for streaming

↓ HTTPS · Cookie: jwt=<token>

API GATEWAY

TRAEFIK · :80

API Gateway

JWT middleware (shared secret with Auth Service)

/api/auth/* → auth:3001 · /api/ai/* → ai:3002

/api/rag/* → rag:3003 · /api/analytics/* → analytics:3004

CORS: localhost:3000

Health checks on all services

JWT payload: { userId, role: "student"|"teacher"|"parent", schoolId, studentId? }

Every service checks role before responding. Parent gets studentId of their child injected at gateway level via parent_student lookup.

↓ Internal Docker network

MICROSERVICES · DOCKER COMPOSE

🐳 DOCKER COMPOSE

:3001 · AUTH_DB SCHEMA

Auth Service

Register · Login · JWT issue

Users · Roles · Schools

parent_student relation table

:3002 · AI_DB SCHEMA

AI Service

Text: → RAG retrieve → Ollama stream

Image: → ComfyUI API

Video: serve pre-generated file

Conversation history in ai_db

Loading SSE event on cold start

:3003 · RAG_DB SCHEMA

RAG Service

Upload PDF → chunk → embed → ChromaDB

/retrieve → top-5 chunks for AI Service

nomic-embed-text via Ollama

512 tok chunks · 50 tok overlap

10 NCERT chapters seeded on boot

:3004 · ANALYTICS_DB SCHEMA

Analytics Service

Scores · Attendance · Query counts

Feedback loop · Intervention flags

Fire-and-forget event POST

Teacher marks attendance via API

Demo data seeded on boot

AI chat request flow: Student → Traefik → AI Service → [RAG /retrieve?q=...](#) (top-5 chunks) → AI Service builds prompt (chunks + history) → [Ollama /api/generate \(stream\)](#) → SSE tokens → browser renders live | AI Service fires **POST analytics/event** (async, no await)

Image request flow: Student → AI Service → [ComfyUI /api/prompt](#) (loads SD on demand, unloads after) → image saved to Docker volume → URL returned to browser

↓ Read / Write

DATA LAYER

POSTGRESQL · :5432

Primary Database

4 isolated schemas

auth_db · ai_db · rag_db · analytics_db

Each service owns + migrates its schema

CHROMADB · :8000

Vector Store

Collections per subject

Semantic search for RAG

Persistent Docker volume (no data loss)

DOCKER VOLUME · /DATA

File Storage

/data/uploads — PDFs

/data/images — generated

/data/videos — pre-generated clips

→ S3 post-funding via ENV swap

↓ HTTP API

LOCAL AI LAYER · ALL ON LAPTOP

OLLAMA · :11434 · ALWAYS LOADED

Qwen2.5 / DeepSeek

Text inference · RAG chat

nomic-embed-text for RAG embeddings

Pulled on first startup via init script

COMFYUI · :8188 · ON DEMAND

Stable Diffusion

Educational image generation

Loads only when image requested

Unloads after — frees VRAM

DOCKER VOLUME · PRE-GENERATED

Video Clips

5 pre-made educational clips

Served from disk instantly

Live gen enabled post-funding

Cloud migration (post-funding): All services read config from ENV vars. No hardcoded localhost anywhere. Ollama URL → Anthropic/OpenAI API · Docker volume → S3 · PostgreSQL → RDS · ChromaDB → Pinecone · Docker Compose → Kubernetes. Each swap is one ENV var change.

- Frontend
- Traefik
- Services
- AI Service
- RAG Service
- Databases
- File Storage
- Ollama (LLM)
- ComfyUI (Image)